

Infosys Springboard Virtual Internship 7.0 Completion Report

Batch :01

Team-C

Start Date :29/06/2026

Team Members:

1. Suryakanth Upadhyay
2. Sanmitha
3. Sabari Vasan

Internship duration – 8 weeks

1. Project Title

Supply Chain Visibility System with Optimization Analytics

Group 1

2. Project Objective

The primary objective of this project is to analyze supply chain operations using the provided dataset to improve visibility across procurement, inventory management, transportation, and delivery processes. The project focuses on identifying operational bottlenecks, evaluating supplier performance, reducing delivery delays, optimizing inventory levels, and improving overall supply chain efficiency through data-driven decision-making.

Additionally, the project aims to identify key performance indicators (KPIs) that influence operational performance and customer satisfaction. By analysing logistics data and creating interactive dashboards, the project enables organizations to monitor business performance effectively and make informed strategic decisions.

This project also aims to demonstrate how data analytics and business intelligence tools can transform raw supply chain data into meaningful business insights. By analyzing various operational parameters such as supplier performance, transportation efficiency, inventory turnover, and delivery time, organizations can identify areas that require improvement and implement strategies to optimize overall supply chain performance.

3. Project Description

The **Supply Chain Visibility & Optimization** project was developed to analyse and improve various aspects of supply chain operations using data analytics and business intelligence techniques. The project focuses on monitoring procurement, inventory management, transportation, warehouse performance, supplier efficiency, customer demand, and delivery operations through interactive Power BI dashboards.

The project began with understanding the business problem and collecting the supply chain dataset provided for the internship. The dataset contained information related to product categories, supplier countries, customer countries, transportation modes, transportation costs, delivery days, inventory turnover, fill rate, stockout orders, and several operational metrics.

Python was used for data preprocessing, where missing values were handled, duplicate records were checked, data types were validated, and unnecessary inconsistencies were removed. After obtaining a clean dataset, it was imported into Microsoft Power BI for visualization and dashboard development.

Five interactive dashboards were created to provide insights into different areas of the supply chain:

- Supply Chain Overview
- Logistic & Inventory Analysis
- Customer & Supplier Analysis
- Delivery & Transportation Analysis
- Warehouse & Inventory Performance

The project demonstrates how business intelligence tools can transform raw operational data into meaningful insights that help organizations improve efficiency, reduce costs, and enhance customer satisfaction.

3.1 Overview

The **Supply Chain Visibility & Optimization** project focuses on analysing supply chain data to understand operational performance and generate meaningful business insights using data visualization. The project provides an overview of different supply chain activities, including transportation, inventory management, supplier performance, customer orders, warehouse operations, and delivery processes. By bringing these aspects together, the project helps provide better visibility into the overall supply chain.

The project uses a structured data analytics workflow that includes problem understanding, data collection, data understanding, data connection, data preprocessing, data modelling, and data visualization. The collected dataset is examined and cleaned to ensure that the information used for analysis is accurate and reliable. Relevant supply chain metrics and Key Performance Indicators (KPIs) are then analysed to identify trends, performance variations, and potential operational bottlenecks.

Microsoft Power BI is used to develop interactive dashboards that present the analysed information through charts, graphs, KPI cards, filters, and other visual elements. These dashboards make complex supply chain data easier to understand and allow users to compare transportation performance, monitor inventory levels, evaluate supplier efficiency, and analyse delivery performance. Interactive filters also enable users to explore the data from different perspectives.

Overall, the project demonstrates how data analytics and visualization can support better supply chain management. The insights obtained from the analysis can help organizations improve operational efficiency, reduce unnecessary costs, monitor performance, and make informed business decisions. The project also provides a foundation for future enhancements such as predictive analytics, demand forecasting, and real-time supply chain monitoring.

3.2 Project Approach

Step 1: Understood the project requirements and business objectives.

Step 2: Collected and studied the supply chain dataset.

Step 3: Cleaned and pre-processed the data using Python by handling missing values and removing inconsistencies.

Step 4: Imported the cleaned dataset into Microsoft Power BI.

Step 5: Created relationships, verified data types, and prepared measures for analysis.

Step 6: Designed interactive dashboards using KPI cards, charts, tables, and filters.

Step 7: Analyzed the dashboards to identify trends, supplier performance, transportation efficiency, inventory status, and delivery performance.

Step 8: Prepared the final report, presentation, and project documentation.

3.3 Technology Used

A combination of data analytics, business intelligence, and collaboration tools was used throughout the project to collect, prepare, analyze, visualize, and manage the supply chain data.

- **Microsoft Power BI Desktop:** Used for importing data, creating the data model, developing interactive dashboards, and presenting supply chain KPIs and visualizations.
- **Power Query:** Used for data cleaning and transformation activities such as removing duplicates, handling missing values, changing data types, and preparing the dataset for analysis.

- **DAX (Data Analysis Expressions):** Used for creating calculated measures and KPIs required for analyzing supply chain performance.
- **Microsoft Excel / CSV:** Used for storing, inspecting, and verifying the supply chain dataset before importing it into Power BI.
- **Kaggle:** Used as the source for obtaining the supply chain dataset used for analysis and visualization.
- **GitHub:** Used for maintaining the project repository, storing project-related files, tracking changes, and managing different versions of the project.
- **GitLab:** Used for project collaboration, repository management, version control, and sharing project resources among team members.
- **Microsoft Teams:** Used for communication, project discussions, mentor interactions, meetings, and coordination during the internship.

3.4 Insights from the Dashboard

The dashboards provide valuable insights into different aspects of supply chain operations. They help analyze transportation patterns, supplier and customer performance, inventory turnover, storage costs, order delays, and delivery performance. By presenting these metrics visually, it becomes easier to identify trends, compare different operational areas, and understand factors affecting overall supply chain efficiency.

The dashboard also helps identify frequently used transportation modes and monitor delayed orders, allowing organizations to evaluate logistics performance and take appropriate corrective actions. Inventory-related visualizations provide information about stock movement and storage, which can support better inventory planning. Overall, these insights help

organizations improve resource utilization, identify operational bottlenecks, and make more informed decisions based on the available data.

3.5 Real-World Impact for Media and Public Communication

The project demonstrates how Business Intelligence and data visualization can be applied to real-world supply chain management. Interactive dashboards convert complex operational information into simple and understandable visual reports, making it easier for managers, stakeholders, and other users to understand supply chain performance and communicate important findings.

The visual reports can support faster decision-making by providing a centralized view of important supply chain metrics. They can also improve coordination between different departments by making relevant information easier to access and share. Better visibility into inventory, transportation, suppliers, and deliveries can contribute to improved logistics planning and customer satisfaction. Thus, the project highlights the practical value of Business Intelligence in communicating data-driven insights and supporting efficient supply chain operations.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project understanding, dataset collection	Understood problem statement and collected dataset
Week 2	Data cleaning	Removed missing values, checked duplicates
Week 3	Data preprocessing	Feature selection and transformation completed
Week 4	Data visualization	Developed initial Power BI dashboards
Week 5	Dashboard enhancement	Added KPIs and interactive visuals
Week 6	Dashboard testing	Validated insights and corrected visuals
Week 7	Analyze dashboard results and generate insights	Analysed delivery, transportation, inventory, supplier, customer, and operational performance
Week 8	Finalize project and documentation	Completed dashboard analysis, documented findings, and prepared the final report

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kick-off	Understood the problem statement, formed the team, collected the dataset, and finalized the project objective.	02/07/2026
Prototype/First Draft	Completed data preprocessing and developed the initial version of the Power BI dashboard.	16/07/2026
Mid-Term Review	Reviewed dashboard visuals, refined KPIs, improved layout, and incorporated mentor feedback.	29/07/2026
Final Submission	Finalized the Power BI dashboard, completed the project documentation, validated the insights, and submitted the completed project.	19/08/2026
Presentation	Presented the completed project, explained the methodology and key insights, and demonstrated the final dashboard.	20/08/2026

Milestone 1 – Data Preparation & First Dashboard

- Understood the supply chain problem and project requirements.
- Collected and examined the 2,000-record, 21-attribute dataset.
- Performed initial data understanding and validation.
- Cleaned the dataset by handling duplicates, missing/inconsistent values, formats, and data types.
- Connected the prepared dataset to Power BI.
- Developed the **first dashboard** as the initial visualization/prototype.

Milestone 2 – Dashboard Expansion

- Continued from the cleaned and connected dataset.
- Improved the Power BI analysis by adding more KPIs and visualizations.
- Developed the **next two dashboards**, expanding the analysis into additional supply-chain dimensions.
- Used visualizations to analyze transportation, inventory, suppliers, delivery, and order-related performance.

Milestone 3 – Advanced Analysis & Insights

- Further developed the dashboard structure and analytical coverage.
- Added the **next two dashboards**.
- Focused more deeply on customer/supplier analysis, delivery and transportation performance, and operational performance.

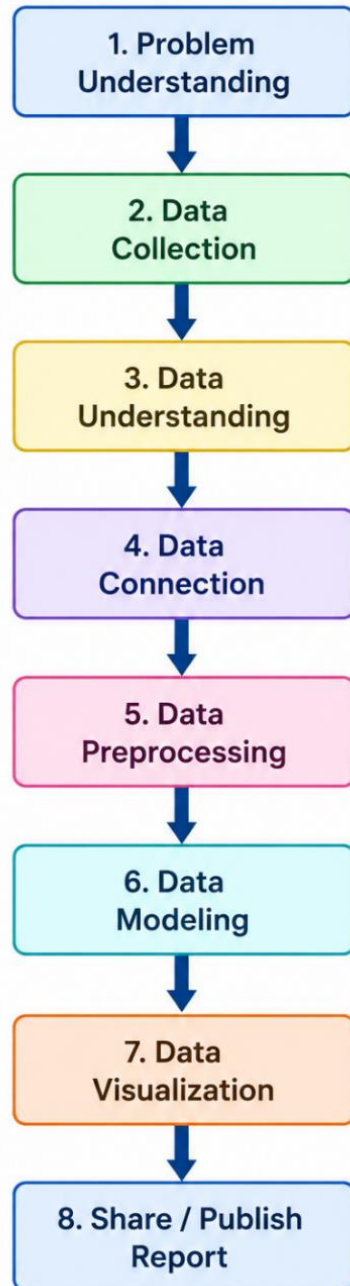
- Compared different supply-chain dimensions to identify trends, delays, costs, and efficiency issues.

Milestone 4 – Final Dashboard & Optimization

- Completed the remaining **two dashboards**, bringing the project to seven dashboards.
- Added more advanced analysis related to **risk, service quality, cost, distance, and inventory efficiency**.
- Reviewed the dashboard outputs and converted the analysis into meaningful supply-chain insights.
- Finalized the project documentation, presentation, and recommendations.

5b. Project execution details

Workflow Diagram



1.Problem Understanding

Problem understanding is the first phase of the project and serves as the foundation for the entire analytical process. Before performing any analysis, it is necessary to understand the business problem, organizational goals, and expected outcomes.

The major questions addressed in this project include:

- Which transportation mode performs the best?
- Which suppliers contribute to delivery delays?
- Are inventory levels maintained efficiently?
- How effectively are customer orders fulfilled?
- Which KPIs influence overall supply chain performance?
- How can operational efficiency be improved using data analytics?

A proper understanding of these business requirements ensures that every subsequent step contributes toward achieving the project objectives.

2.Data Collection

Data collection is the process of gathering relevant data required for analysis. The success of any data analytics project depends on the quality, completeness, and accuracy of the collected data. In this project, the supply chain dataset was provided in CSV format and contains information related to procurement, suppliers, inventory, transportation, and customer deliveries.

The dataset includes various attributes such as product category, supplier country, customer country, transportation mode, transport cost, delivery days, lead time, inventory turnover, fill rate, stockout indicator, and supplier performance. These attributes provide a comprehensive view of the organization's supply chain operations and serve as the foundation for further analysis.

During the data collection stage, the dataset was verified to ensure that all required information was available for the project objectives. The collected data was then stored in a structured format, making it easier to import into Python for preprocessing and later connect to Power BI for visualization. A well-organized and complete dataset ensures accurate analysis, reliable insights, and effective decision-making throughout the project.

3.Data Understanding

After collecting the dataset, the next step is to understand its overall structure and characteristics. This stage helps identify the nature of the available data before any processing begins.

The dataset is examined to determine:

- Number of rows and columns
- Data types of each attribute
- Numerical and categorical variables
- Missing values
- Duplicate records

- Statistical summaries
- Data distributions

Exploratory Data Analysis (EDA) is also performed to understand relationships among different variables and identify patterns within the data. Various statistical measures such as mean, median, standard deviation, minimum, and maximum values are calculated.

Understanding the dataset helps identify inconsistencies, detect anomalies, and determine which variables are most useful for achieving the project objectives.

4.Data Connection

Data connection refers to importing the collected dataset into analytical tools for further processing and visualization.

In this project, the CSV dataset is imported using the Pandas library in Python. After importing the dataset, several validation checks are performed to ensure that all records have been loaded correctly without any errors.

The following activities are carried out:

- Import the CSV file.
- Load data into a Pandas Data Frame.
- Verify dataset dimensions.
- Check column names.
- Validate data types.

- Generate summary statistics.
- Ensure successful loading of records.

Once the dataset is validated, it can also be connected to Power BI for dashboard development. Establishing a reliable data connection ensures consistency throughout the analysis and enables seamless integration between Python and Power BI for visualization and reporting.

5.Data Preprocessing

Data preprocessing is one of the most important stages in the data analytics workflow because real-world datasets often contain missing values, duplicate records, inconsistencies, and formatting issues.

To improve data quality, the following preprocessing steps were performed:

- Checked for missing values.
- Removed duplicate records.
- Standardized column names.
- Verified numerical and categorical data types.
- Identified outliers using descriptive statistics.
- Corrected inconsistent values.
- Selected relevant features for analysis.
- Organized data into a structured format.
- Validated data consistency.
- Prepared the final clean dataset for visualization.

A properly cleaned dataset reduces analytical errors, improves model accuracy, and ensures reliable business insights. Data preprocessing significantly enhances the quality and credibility of the overall analysis.

6.Data Modelling

After preprocessing, the cleaned dataset is used for data modelling and KPI analysis. Data modelling involves transforming raw operational data into meaningful business metrics that help evaluate organizational performance.

Several Key Performance Indicators (KPIs) are calculated during this stage, including:

- Average Delivery Time
- Transportation Cost
- Fill Rate
- Inventory Turnover
- Lead Time Variability
- Supplier Performance
- Stockout Rate
- Order Fulfilment Efficiency

These KPIs help organizations monitor operational performance, identify inefficiencies, compare suppliers, evaluate logistics operations, and improve business decision-making.

Data modelling converts raw transactional data into actionable business information that supports strategic planning and continuous improvement.

7.Data Visualization

After completing data modelling and KPI analysis, the next stage is to present the analysed data through meaningful visualizations. In this project, Microsoft Power BI is used to create interactive dashboards that provide a clear overview of supply chain performance. Different charts, graphs, KPI cards, filters, and slicers are used to represent important operational metrics in an easy-to-understand format.

The dashboard visualizations are used to analyze transportation performance, supplier efficiency, inventory levels, delivery time, transportation costs, and order fulfilment. Interactive filters allow users to explore the data based on different categories and identify trends or areas that require attention. These visualizations make complex supply chain data easier to interpret and help users compare different operational factors.

The main objective of data visualization is to convert the results of the analysis into actionable insights. The Power BI dashboard helps identify operational bottlenecks, monitor important KPIs, and support data-driven decision-making. Effective visualization also improves communication of analytical findings to managers and other stakeholders.

Page 1: Supply Chain Overview

Visual Explanations

1. Total Orders (Card)

- Displays the total number of customer orders in the dataset.
- This KPI provides a quick overview of business activity.

2. Average Delivery Days (Card)

- Shows the average number of days required to deliver an order.
- It helps evaluate delivery efficiency and customer service performance.

3. Total Transportation Cost (Card)

- Displays the total transportation cost incurred for all shipments.
- Useful for monitoring logistics expenditure.

4. Delay Status (Donut Chart)

- Compares delayed orders and on-time deliveries.
- Helps identify the percentage of delayed shipments.

5. Orders by Product Category (Column Chart)

- Compares the number of orders across different product categories.
- Identifies the most demanded products.

6. Transportation Cost by Transport Mode (Bar Chart)

- Displays transportation cost for Air, Road, Rail, and Sea transport.

- Helps determine which transport mode contributes the highest logistics cost.

7. Customer Distribution (Tree map)

- Shows the number of customer orders received from different countries.

Larger blocks indicate countries contributing more customer orders.

Page 2: Logistic and Inventory Analysis

Visual Explanations

1. Total Stockout Orders (Card)

- Displays the number of stockout orders.
- Indicates situations where products were unavailable to fulfill customer demand.

2. Average Fill Rate (%) (Card)

- Shows the average percentage of customer demand fulfilled successfully.
- Higher fill rate indicates efficient inventory management.

3. Average Cost Per Kilometre (Card)

- Displays the average transportation cost per kilometre.
- Helps measure transportation efficiency.

4. Transportation Cost by Supplier Country (Bar Chart)

- Compares transportation costs for suppliers from different countries.
- Helps identify suppliers with higher logistics expenses.

5. Delayed Orders by Transport Mode (Stacked Column Chart)

- Shows delayed and non-delayed orders for each transport mode.
- Helps identify transport modes experiencing more delays.

6. Inventory Turnover by Product Category (Column Chart)

- Compares inventory turnover among product categories.
- Higher turnover indicates faster inventory movement.

7. Average Storage Cost by Product Category (Bar Chart)

- Displays average storage cost for each product category.
- Helps identify products requiring higher warehouse storage costs.

Page 3: Customer & Supplier Analysis**Visual Explanations****1. Total Supplier Countries (Card)**

- Displays the total number of supplier countries.
- Indicates supplier network coverage.

2. Total Customer Countries (Card)

- Displays the number of customer countries.
- Represents the geographical reach of customers.

3. Quantity by Customer Country (Column Chart)

- Compares product quantity ordered by customers from different countries.
- Helps identify major customer markets.

4. Average Delivery Days by Supplier Country (Bar Chart)

- Shows average delivery time for suppliers from different countries.
- Helps evaluate supplier efficiency.

5. Orders by Season (Donut Chart)

- Displays order distribution across Summer, Winter, Spring, and Autumn.
- Helps analyze seasonal demand.

6. Orders by Supplier Country (Bar Chart)

- Compares the number of orders received from suppliers in different countries.
- Identifies suppliers with higher order volumes.

7. Quantity by Transport Mode (Column Chart)

- Displays product quantity transported through different transportation modes.
- Helps understand transport mode utilization.

Page 4: Delivery & Transportation Analysis

Visual Explanations

1. Total Delayed Orders (Card)

- Displays the total number of delayed orders.
- Helps monitor delivery performance.

2. Total On-Time Orders (Card)

- Displays the number of successfully delivered orders.
- Indicates delivery reliability.

3. Average Transport Days (Card)

- Shows the average number of transport days.
- Helps evaluate logistics efficiency.

4. Orders by Transport Mode (Donut Chart)

- Displays the distribution of orders across Air, Road, Rail, and Sea transport.
- Helps understand transport mode preference.

5. Delivery Days by Season (Bar Chart)

- Compares average delivery days across different seasons.
- Helps analyze seasonal delivery performance.

6. Delay Days by Product Category (Column Chart)

- Displays average delay days for each product category.
- Helps identify products affected by longer delays.

7. Transport Days by Mode (Column Chart)

- Compares average transport days among transportation modes.
- Helps identify the fastest and slowest transport options.

Page 5: Warehouse & Inventory Performance

Visual Explanations

1. Total Quantity Order (Card)

- Displays the total quantity of products ordered.
- Indicates overall demand volume.

2. Average Order Processing Days (Card)

- Shows the average time required to process customer orders.
- Helps evaluate operational efficiency.

3. Total Delay Days (Card)

- Displays the total delay days recorded across all orders.
- Indicates the overall delay impact.

4. Transportation Cost by Season (Area Chart)

- Shows transportation cost variation across different seasons.
- Helps identify seasonal logistics expenses.

5. Quantity by Supplier Country (Funnel Chart)

- Displays the quantity supplied by each supplier country.
- Helps compare supplier contributions.

6. Order Processing Days by Product Category (Bar Chart)

- Compares average processing days across product categories.
- Helps identify products requiring more processing time.

7. Quantity by Season (Area Chart)

- Displays product quantity ordered during different seasons.
- Helps understand seasonal demand patterns.

Page 6 – Supply Chain Risk & Service Quality

1. Average Transportation Distance

This KPI shows the **average distance travelled per order**, which is **7.72K km**. It helps measure the transportation scale of the supply chain and understand the typical distance involved in delivering products.

2. Average Quantity per Order

This KPI shows an average order quantity of **492.88 units**. It indicates the approximate volume handled in each order and helps understand the overall order size.

3. Average Delay Days

The average delay is **1.10 days**. This provides an indication of delivery reliability and helps identify whether delays are a significant operational issue.

4. Delivery Reliability by Transport Mode

This stacked bar chart compares **on-time and delayed orders across transport modes**. Sea has the highest number of orders and also the highest number of delayed orders, while air and rail have fewer orders. This helps identify which transportation modes require greater attention for improving delivery reliability.

5. Average Fill Rate by Transport Mode

This chart compares the average **fill rate percentage** for different transport modes. The values are around **93–94%**, showing that most customer demand is being fulfilled successfully across all transportation modes.

6. Total Delay Days by Supplier Country

This chart shows the total delay days contributed by each supplier country. The **USA has the highest total delay days (328)**, followed by India and Japan. This helps identify supplier locations where delivery delays may need closer monitoring.

7. Stockout Analysis by Product Category

This stacked column chart compares stockout and non-stockout orders across product categories. **Pharma has the highest number of orders**, while

Clothing has the lowest. The chart helps identify product categories where stock availability should be monitored to reduce stockout situations.

Page 7 – Cost, Distance & Inventory Efficiency

1. Maximum Delivery Days

This KPI shows the **maximum delivery time as 74 days**. It highlights the longest delivery duration observed in the dataset and helps identify potential extreme delivery cases.

2. Average Transport Cost per Order

The average transportation cost per order is **8.60K USD**. This KPI helps evaluate the typical transportation expenditure associated with fulfilling an order.

3. Orders by Season

This column chart compares the number of orders across seasons. **Summer has the highest number of orders (531)**, followed by winter, spring and autumn. It helps identify seasonal variations in demand.

4. Average Inventory Turnover by Season

This line chart shows inventory turnover across seasons. **Winter has the highest turnover (8.21)**, while spring has the lowest (**8.00**). Higher turnover indicates that inventory is being utilized or replenished more frequently.

5. Average Transport Cost by Product Category and Transport Mode

These visual compares transportation costs across **product categories and transport modes**. It helps identify how transportation expenses vary

depending on both the type of product and the transportation method, supporting better cost-management decisions.

6. Delivery Time Composition

The waterfall chart shows how the average transportation time is composed across different transport modes. Sea contributes **57.91 days**, rail **52.96 days**, road **50.42 days**, and air **14.42 days**, resulting in a total of approximately **175.71 days**. This helps understand the contribution of each mode to the overall transportation time.

7. Quantity by Product Category

This bar chart compares the total quantity across product categories. **Pharma has the highest quantity at approximately 0.21M**, while Clothing has the lowest at approximately **0.18M**. This helps identify which product categories contribute the greatest volume.

8.Publish Report

The final stage of the workflow is to share and publish the completed dashboard and project report. After completing the data analysis and visualization, the results are organized into a structured report containing the project objectives, data analysis process, preprocessing activities, KPI analysis, dashboard visualizations, insights, and recommendations.

The Power BI dashboard and final report can be shared with project stakeholders and team members for review and discussion. Publishing the results makes the analyzed information accessible to users and allows them to understand the overall supply chain performance. The report also provides a documented summary of the complete analytical process followed during the project.

Sharing the final results ensures that the insights obtained from the analysis can be used for better planning, performance monitoring, and decision-making. This completes the data analytics workflow by transforming the collected supply chain data into useful information that can support operational improvement.

6. Screenshots

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IMAGE 1

6.1 DASHBOARD 1 – SUPPLY CHAIN OVERVIEW

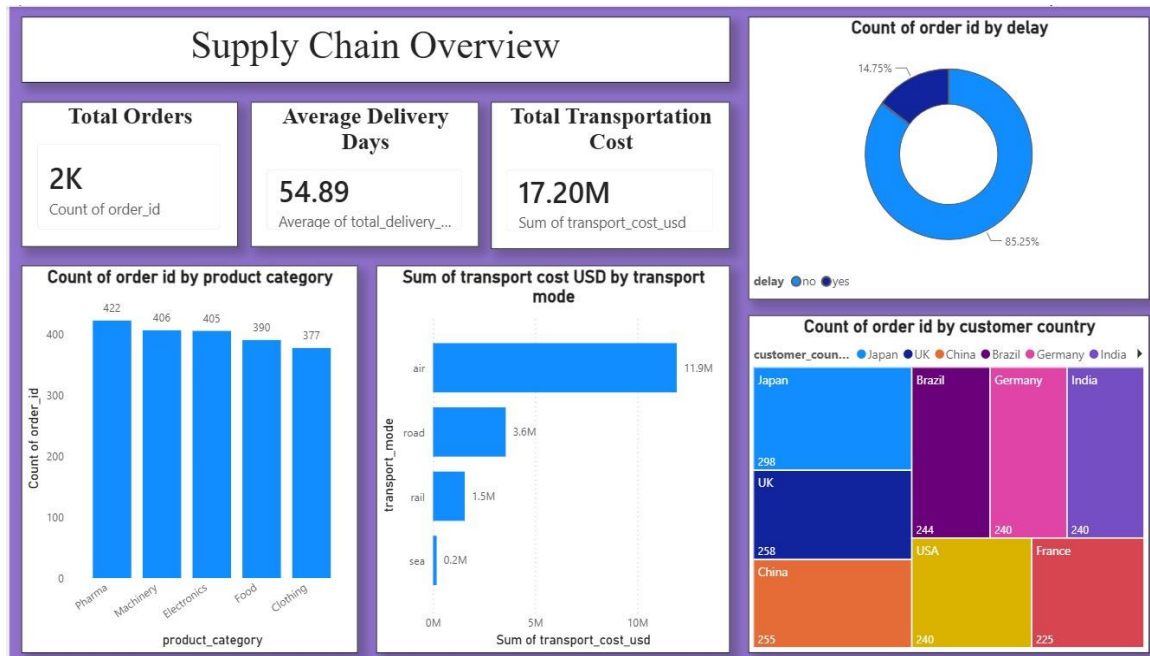


IMAGE 2.1

6.2 DASHBOARD 2- LOGISTIC AND INVENTORY ANALYSIS

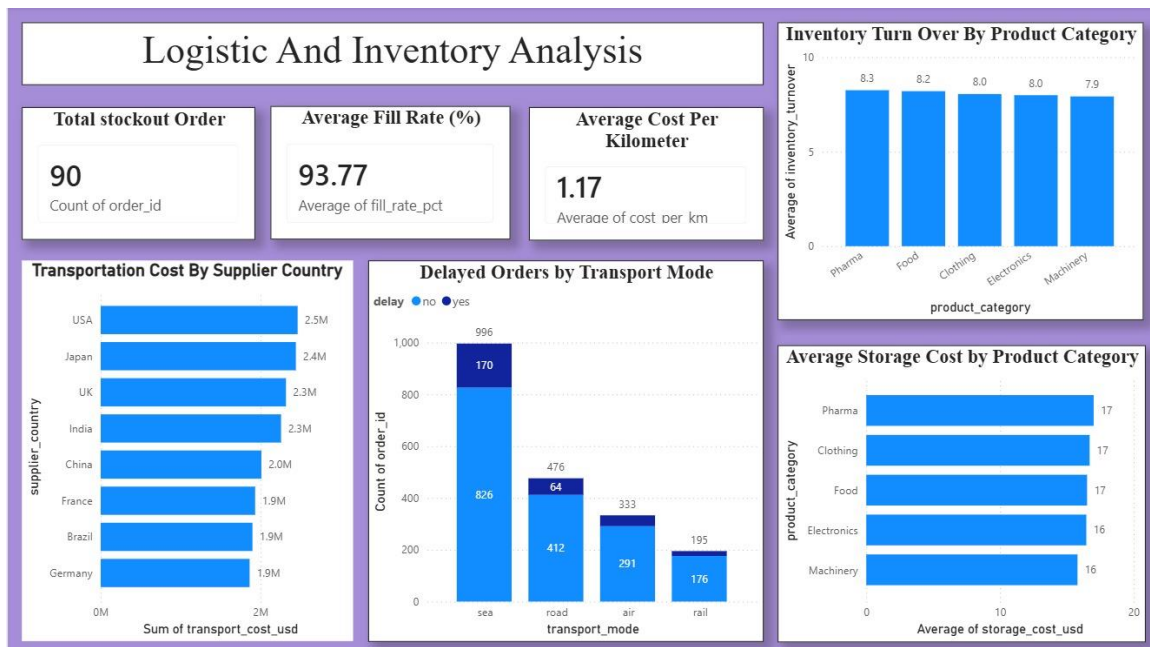


IMAGE 2.2

6.3 DASHBOARD 3- CUSTOMER & SUPPLIER ANALYSIS



IMAGE 2.3

6.4 - DASHBOARD 4 – DELIVERY AND TRANSPORTATION ANALYSIS

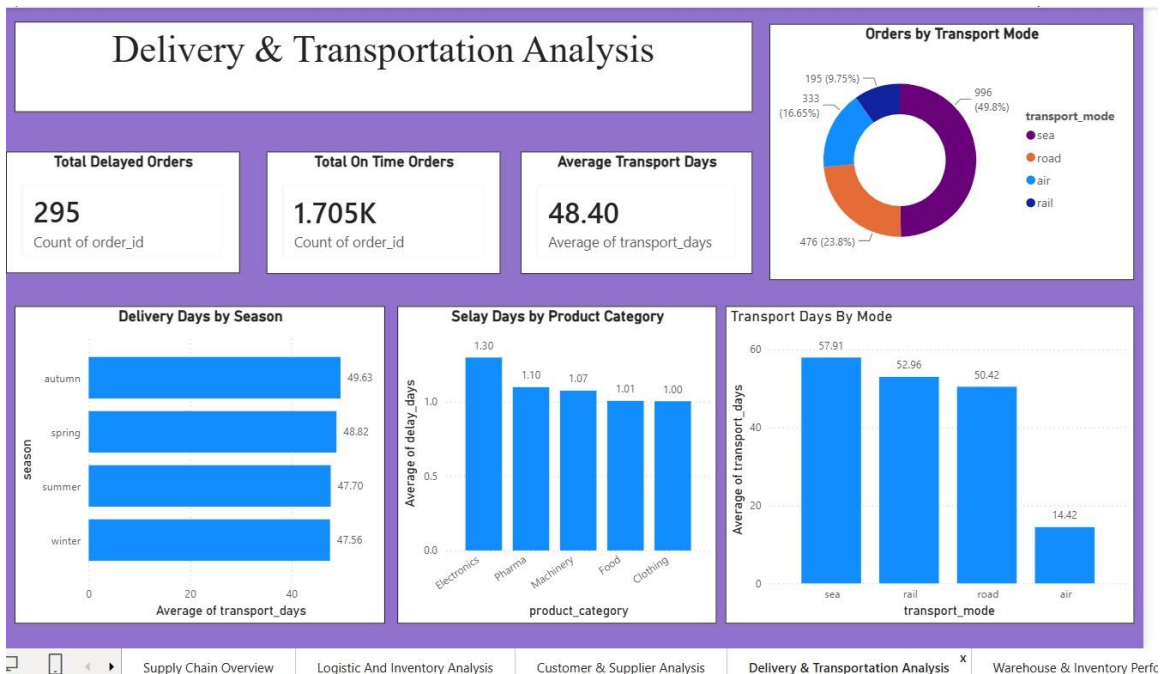


IMAGE 2.4

6.5 -DASHBOARD 5 – OPERATIONAL PERFORMANCE DASHBOARD

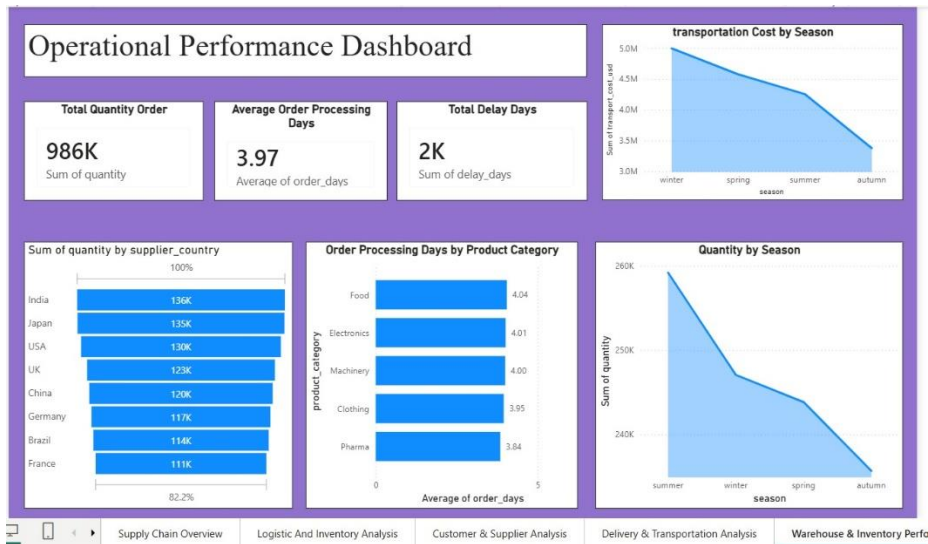


IMAGE 2.5

6.6-DASHBOARD 6 – SUPPLY CHAIN RISK & SERVICE QUALITY



IMAGE 2.6

7-DASHBOARD 7 – COST DISTANCE & INVENTORY EFFICIENCY

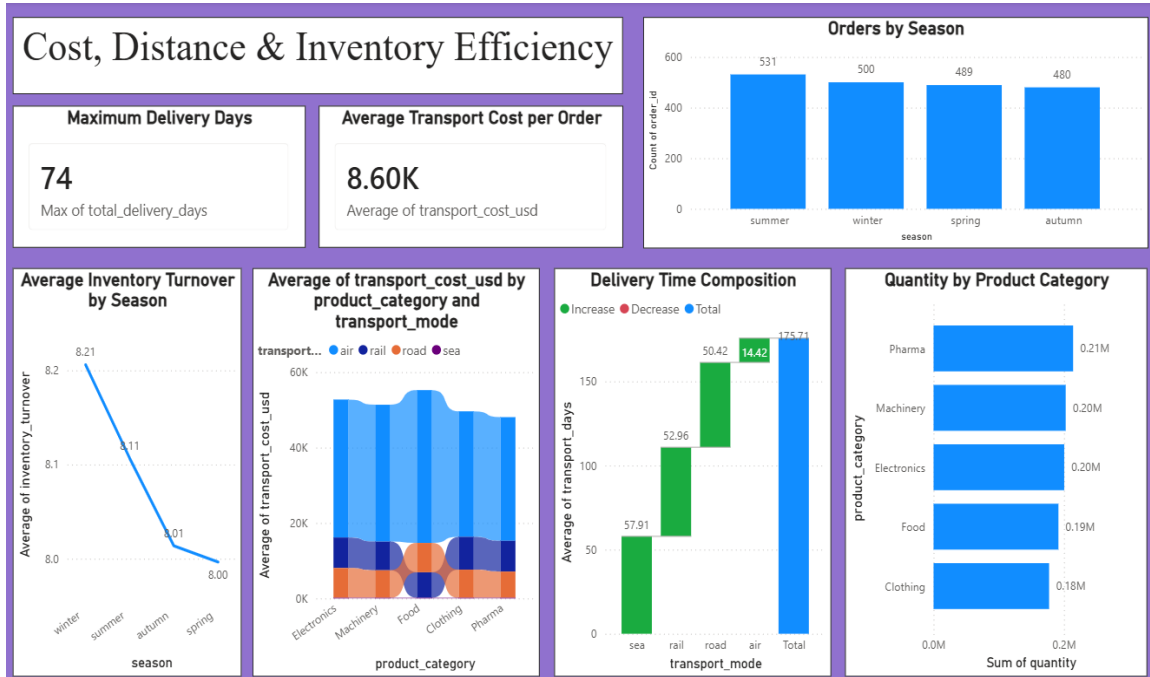


IMAGE 2.7

7. Challenges Faced

During the internship, one of the initial challenges was **understanding the supply chain dataset and identifying the most relevant KPIs** for the project. The dataset contained different types of information related to orders, products, customers, suppliers, transportation, delivery and inventory. Understanding the purpose of each field and determining which metrics were useful for business analysis required careful study of the dataset.

Another challenge was **data preprocessing and validation**. The data had to be checked for missing values, inconsistent records and unsuitable data formats before creating the dashboards. Proper preprocessing was important because inaccurate or incomplete data could lead to misleading visualizations and incorrect business insights. The data was therefore reviewed and validated before being used for analysis.

Selecting appropriate Power BI visualizations was another challenge faced during the project. Different types of charts were experimented with depending on the type of information being presented. KPI cards were used for important summary values, while bar charts, column charts, line charts, stacked charts and other visuals were used to compare categories and identify trends. The main focus was to make every visualization meaningful and easy to interpret.

Another difficulty was **avoiding repetition across multiple dashboard pages**. Since several pages were created for different areas of supply chain analysis, it was important to ensure that each page provided different and useful information. Visuals were selected based on specific analytical

objectives such as delivery performance, transportation, inventory, supplier performance, customer analysis and operational efficiency.

Dashboard layout and formatting also required considerable attention. The placement of charts, KPI cards, tables and headings had to be planned carefully so that the dashboards were clear and visually balanced. A consistent background, font style, formatting and overall theme were maintained across the pages to give the complete dashboard a professional appearance.

8. Learnings & Skills Acquired

This internship provided our team with practical experience in data analytics and business intelligence, particularly in applying a structured workflow to a real-world supply chain analysis problem.

As a team, we gained hands-on knowledge in data collection, data understanding, exploratory data analysis, data connection, data preprocessing, data modeling, KPI analysis, and data visualization. We worked with Python, Pandas, and Microsoft Power BI for data validation, analysis, transformation, and interactive dashboard development.

The project enhanced our ability to identify meaningful KPIs, select appropriate visualizations, interpret business data, generate actionable insights, and present analytical findings effectively. Along with technical knowledge, the internship strengthened our teamwork, problem-solving, analytical thinking, documentation, project management, communication, and presentation skills. Overall, the experience helped us understand how data-

driven techniques can be applied collaboratively to address real-world business challenges.

8.1 Technical Skills

- Learned data preprocessing.
- Developed interactive dashboards using Microsoft Power BI.
- Gained experience in creating KPI cards, charts, tables, and data visualizations.
- Improved knowledge of data analysis and dashboard design.

8.2 Analytical and Problem-Solving Skills

- Improved ability to analyze large datasets.
- Learned to identify trends, patterns, and business insights.
- Developed problem-solving skills by interpreting data and suggesting improvements.

8.3 Future Enhancements

- Integrate real-time supply chain data.
- Add predictive analytics using Machine Learning.
- Develop automated alerts for delays and stock shortages.
- Publish dashboards online for remote monitoring and collaboration.

8.4 Soft Skills and Team Collaboration

- Improved communication and presentation skills.
- Enhanced teamwork through collaborative project development.
- Learned effective time management and task planning.
- Gained confidence in preparing professional reports and presentations.

8.5 Domain Knowledge and Application

This project improved my understanding of supply chain management concepts such as inventory control, transportation management, warehouse operations, supplier evaluation, customer demand analysis, and logistics optimization. It also demonstrated how data analytics and business intelligence can be applied to solve real-world business problems and support strategic decision-making.

9. Testimonials from Team

Suryakant Upadhyay:

The internship provided an opportunity to gain practical knowledge of supply chain analytics and data visualization. Working collaboratively with the team helped in understanding different perspectives, solving project challenges, and improving decision-making skills. The experience also strengthened technical abilities in data handling and visualization while providing valuable exposure to real-world industry

Sanmitha:

The internship was a valuable learning experience that provided practical exposure to **data analysis and supply chain management**. Working with the team helped in developing better communication, collaboration, and problem-solving skills. The experience of analyzing real-world data and contributing to the project improved both technical knowledge and confidence in handling business intelligence tasks.

Sabari Vasan N:

The internship provided valuable hands-on experience in **supply chain analytics and business intelligence**. Working on data preprocessing, analysis, and Power BI dashboard development helped in strengthening analytical and technical skills. Collaborating with team members also improved communication, coordination, and problem-solving abilities while working toward a common project goal.

10. Conclusion

This project demonstrates how data analytics and business intelligence techniques can significantly improve supply chain visibility and operational efficiency. By understanding business requirements, collecting and preprocessing data, analyzing key performance indicators, and developing interactive Power BI dashboards, organizations can gain valuable insights into procurement, inventory, transportation, and delivery operations.

The project highlights the importance of data-driven decision-making in reducing operational costs, minimizing delivery delays, improving supplier performance, and enhancing customer satisfaction. The dashboards developed during this project provide decision-makers with an effective tool for monitoring supply chain performance and identifying opportunities for continuous improvement.

In the future, this project can be further enhanced by integrating machine learning models for demand forecasting, delivery delay prediction, route optimization, and predictive inventory management. Such advanced analytical techniques can help organizations build smarter, more resilient, and highly efficient supply chain systems.

11. Acknowledgement

We would like to express our sincere gratitude to the Infosys Springboard Team for organizing this highly enriching virtual internship program and for providing a structured, industry-aligned learning platform. The well-designed modules, milestone-based framework, and practical orientation of the internship enabled us to gain meaningful exposure to real-world business intelligence applications.

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internship has been a meaningful milestone in our academic and professional journey, reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.